

Microwave Imaging Framework

Beginner-Friendly Guide

What happens in each module, what you see in the GUI, and how to read a real sample result

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Written for new students — plain words, correct flow, real sample output

Important: App output is **not a clinical diagnosis**. It is a research / learning tool under stated assumptions.

1. The big picture (in one page)

Think of the app as a factory line. You put microwave measurement files in at the start. Each station (module) cleans the signals, turns them into an image, finds a bright suspicious spot, measures it, and writes a report. You mostly click through three big tabs: **Acquisition** → **Preprocessing** → **Reconstruction**.

Step	Tab you use	What goes in	What comes out
1 Load	Acquisition	.mat / .s2p / BMID cube	A checked dataset ready to use
2 Clean	Preprocessing	Dataset	Cleaner signals (or unchanged for BMID)
3 Image	Reconstruction	Signals + antenna settings	Breast/phantom image + ROI + scores
4 Save	Report / toolbar	Everything above	Markdown + pictures under results/

Two kinds of data (do not mix their recipes):

- **UM-BMID (.mat cube)**: already cleaned. Use preset **BMID pass-through** (leave cleaning OFF).
- **UG Touchstone (.s2p)**: needs cleaning. Use preset **.s2p defaults** (mild filter + Week 3).

2. Tour of the GUI (what every screen is for)

2.1 Main window

- Window title: **Microwave Imaging Framework**.
- Three tabs: **Acquisition**, **Preprocessing**, **Reconstruction**.
- Top toolbar: **Download Final Report** + a status line like *Auto-report: waiting for pipeline....*
- After preprocess/reconstruct, the app quietly saves `results/latest_session_report.md`.

2.2 Acquisition tab

- Title: **Data Acquisition**.
- Buttons: **Load dataset** and **Run Auto Validation**.
- Left: **Dataset** table (Field / Value) — file name, frequencies, ports, BMID tumor info.
- Right: **Log** — step-by-step messages.
- BMID cubes open a dialog **Select UM-BMID Scan** with filter All / Tumor only / Healthy only.
- After a good load, the app jumps you to Preprocessing.

2.3 Preprocessing tab

- Buttons: **BMID pass-through**, **.s2p defaults**, **Run**, **Export**.
- Inner tabs: **Plots** | **Settings** | **Results**.
- Settings = cleaning stages (filter, calibration, normalization, artifacts) + Touchstone options (spike method, Week 3).
- After Run: Plots update (before/after signals). For .s2p you may also see phase unwrap and time-domain.
- Then the app switches to Reconstruction automatically.

2.4 Reconstruction tab

- Buttons: **Run**, **Auto Tweak**, **Report**.
- Inner tabs: **Image** | **Settings** | **Details** | **Compare** | **ROI refine**.
- Settings: Geometry (radius, FOV, wave speed, flips...) and Imaging (ROI mode, beamformer pick).
- BMID tip shown in UI: radius 18 cm, FOV 12×12 cm, $c = 3.0e8$, Peak ROI, Prefer DMAS-D4.
- After Run, Image tab shows the selected reconstruction with cyan ROI box, white centroid, lime Tumor GT × when available.

3. Each module: input → inside process → output → GUI view

Module 1 — Load & check data

Input: A file path (.mat / .s2p) or BMID cube + scan choice.

Inside (simple): Open file → check frequencies and S-parameters are finite and ordered → read metadata (tumor/healthy, antenna size).

Output: A MicrowaveDataset object + summary numbers.

Where you see it in the GUI: Acquisition Dataset table fills; Log says loaded successfully; app moves to Preprocessing.

Module 2 — Clean signals

Input: The loaded dataset + Settings (or a preset).

Inside (simple): For .s2p: spike fix, mild filter, optional Hamming/IFFT/reference/clutter. For BMID _adi: do almost nothing (pass-through).

Output: Processed signals + quality numbers (SNR before/after).

Where you see it in the GUI: Plots tab shows before/after curves; Results summary updates; Reconstruction becomes ready.

Module 3 — Frequency to time

Input: Frequency-domain S21.

Inside (simple): IFFT turns frequency samples into a time signal (like converting a recipe into a timeline).

Output: Time-domain traces used by beamformers / Week-3 plots.

Where you see it in the GUI: On .s2p with Week 3 enabled: Plots may show a time-domain panel.

Module 4 — Make the image

Input: Signals + geometry (antenna circle, FOV, wave speed).

Inside (simple): Three algorithms draw an image of where energy focuses: DAS, DMAS, DMAS-D4.

Output: Three 2D images on the same grid.

Where you see it in the GUI: Reconstruction Image + Compare tabs.

Module 5 — Pick the best image

Input: The three images + quality scores.

Inside (simple): Compute SNR/SCR/contrast; choose by Quality / Prefer DMAS-D4 / Force... / Closest to tumor GT.

Output: One selected beamformer name + score table.

Where you see it in the GUI: Details Summary + Compare metrics table.

Module 6 — Find the hot spot (ROI)

Input: Selected image.

Inside (simple): Find bright connected region; score prefers compact foci; may avoid huge edge clutter.

Output: Bounding box, centroid, area, ROI score.

Where you see it in the GUI: Cyan rectangle on Image; ROI table in Details.

Module 7 — Zoom & re-image ROI

Input: ROI box + original signals.

Inside (simple): Rebuild only that small area on a finer grid.

Output: Sharper local image (ROI refine).

Where you see it in the GUI: ROI refine tab plot + ROI details table.

Module 8 — Tumor candidate decision

Input: ROI features (size, off-center-ness, brightness).

Inside (simple): Compute suspicion; penalize huge edge blobs; compare to threshold (~0.45).

Output: Yes/No + candidate confidence + short reason.

Where you see it in the GUI: Details ROI rows: Tumor Candidate, Candidate Confidence, Suspicion Score.

Module 9 — Measure the region

Input: ROI + physical FOV size.

Inside (simple): Convert pixels to cm: diameter, bbox, area, FWHM, compactness, intensities.

Output: A list of physical measurements.

Where you see it in the GUI: Details ROI rows: Centroid (cm), Eq. Diameter, BBox, Area, FWHM, Local SCR...

Module 10 — Overall confidence

Input: Image quality + detection + beamformer margin + localization + focus.

Inside (simple): Blend into one 0–1 score and label low/medium/high with short notes.

Output: Overall confidence number + component breakdown.

Where you see it in the GUI: Details: Overall Confidence, Conf: Detection, Conf: Localization; also in session report.

Module 11 — Visualization

Input: All pipeline results.

Inside (simple): Draw plots and tables so humans can check the work.

Output: Interactive views.

Where you see it in the GUI: Everything you see in Plots / Image / Compare / ROI refine.

Module 12 — Reports

Input: Dataset + preprocess + reconstruction snapshot.

Inside (simple): Write Markdown + JSON + PNG figures; optional folder copy.

Output: Session report package under results/.

Where you see it in the GUI: Toolbar Download Final Report / Reconstruction Report button; Auto Validation writes guide+beginner MD.

4. Practical click checklist (BMID demo)

- Put `fd_data_s21_adi.mat` + `md_list_s21_adi.mat` in `project/datasets/`.
- Start app: `python main.py`.
- Acquisition → Load dataset → pick a tumor scan in the BMID dialog.
- Preprocessing → confirm **BMID pass-through** → Run.
- Reconstruction → check tip settings (18 cm, 12×12 cm, $c=3e8$, Peak ROI, Prefer/Force DMAS-D4) → Run.
- Look at Image (ROI box + GT mark), Details (candidate/confidence), Compare (three algorithms), ROI refine.
- Click Report (or toolbar Download Final Report) to save under results/.
- Optional: Acquisition → Run Auto Validation for batch screening + dual reports.

5. Reading the Details table (cheat sheet)

GUI field	Plain meaning
Selected Beamformer	Which algorithm won (often DMAS-D4 on BMID)
Tumor Candidate Yes/No	Does the hot spot look tumor-like vs clutter?
Candidate Confidence	How strong that Yes/No evidence is (0–1)
Eq. Diameter (cm)	Rough size of the region if treated as a circle
FWHM (cm)	How wide the bright peak is (large = blurry/diffuse)
Overall Confidence	Combined reliability (low/medium/high) — not a diagnosis
Localization error	Distance from ROI centroid to known tumor mark (if metadata has GT)

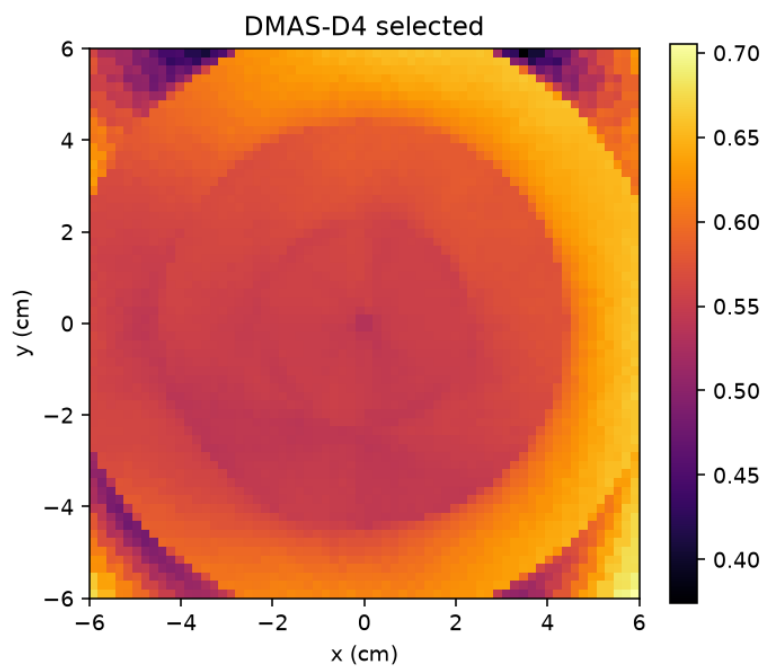
6. Final sample GUI output (real run from this project)

Below is a real BMID sample that was run in this workspace (scan 0 from `fd_data_s21_adi.mat`, phantom A2F1, labeled tumor). Preprocessing was pass-through. Reconstruction selected **DMAS-D4**. These figures are the same kind of pictures the GUI shows on Image / Compare / ROI refine tabs.

6.1 What was loaded

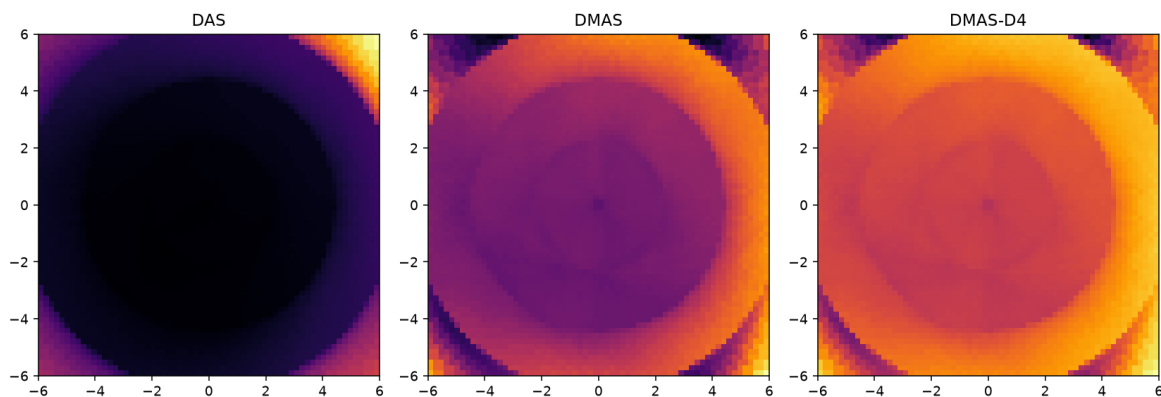
Item	Value
File	fd_data_s21_adi.mat (UM-BMID)
Scan	0 / id 1 / phantom A2F1 / tumor = True
Frequencies	1001 points from 1–8 GHz
Traces	72
Tumor GT (metadata)	3.0 cm diameter at (2.25, 2.25) cm
Preprocess	all none (pass-through)
Selected beamformer	DMAS-D4 (force_dmas_d4)
Grid / FOV / radius / c	64×64 / 12×12 cm / 18 cm / 3e8 m/s

6.2 Selected reconstruction (GUI Image tab)



Selected reconstruction (DMAS-D4). In the live GUI you also see a cyan ROI box, white centroid, and a lime x for Tumor GT when metadata exists.

6.3 Beamformer comparison (GUI Compare tab)



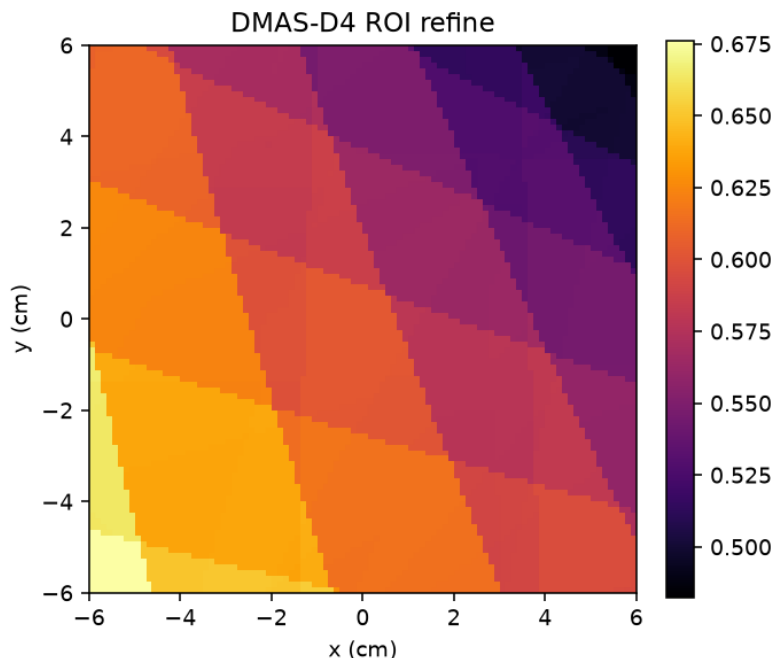
DAS vs DMAS vs DMAS-D4 side by side — same experiment, three algorithms.

6.4 Quality scores (what Compare/Details summarize)

Algorithm	Score	SNR	SCR	Contrast
DAS	0.38	1.30	12.35	0.86
DMAS	0.26	3.60	9.36	0.85
DMAS-D4 (selected)	0.70	14.64	22.12	0.31

Plain reading: DMAS-D4 had the highest combined score and much higher SNR/SCR, so the app preferred it. Higher SCR usually means a clearer peak versus background clutter.

6.5 ROI refine view



High-resolution re-image of the ROI region (GUI: ROI refine tab).

6.6 Detection / characterization / confidence (GUI Details → ROI)

Result	Sample value	Simple meaning
Tumor Candidate	Yes	Features looked more tumor-like than sheet clutter
Candidate confidence	0.91	Strong candidate evidence
Suspicion score	1.14	Raw tumor-likelihood feature score
Eq. diameter	0.83 cm	Compact region size estimate
ROI centroid	(-5.74, -5.74) cm	Where the ROI peak sits in the FOV
Tumor GT	(2.25, 2.25) cm	Where metadata says the tumor is
Localization error	11.30 cm	ROI is far from GT — geometry likely wrong
Overall confidence	0.61 (medium)	Detection strong, but localization weak

7. Interpretation of this sample (honest reading)

7.1 What the pictures suggest

- DMAS-D4 produced a structured energy map with a compact bright region the detector marked as a candidate.
- Candidate Yes + high candidate confidence means the local pattern looked off-center/compact rather than a huge edge sheet.
- However, the ROI centroid is near a corner (~ -5.7 cm, -5.7 cm) while tumor GT is at $(+2.25, +2.25)$ cm — about **11.3 cm** away.

7.2 What that means for trust

This is a classic “pretty image, wrong place” case. Image quality metrics (SCR/SNR) and tumor-candidate features can look good even when the antenna geometry (angle offset / flip / arc) is misaligned with the true phantom. That is why Overall Confidence is only **medium (0.61)**: detection components are high, but localization component is very low.

7.3 What you should try next in the GUI

- Keep BMID pass-through and $c = 3e8$.
- Try Angle offset $90^\circ/180^\circ/270^\circ$, Rotation CW, Axis flip X/Y, Antenna arc 355° (BMID).
- Or click **Auto Tweak** to search common geometry combinations.
- Watch localization error in the session report / Details — aim for a few centimeters, not >10 cm.
- Compare a healthy scan with the same settings to see if the candidate disappears.

7.4 Truthfulness statement

Projected hotspots are algorithm focuses under assumptions, not medical findings. Confidence scores are heuristic research signals. **Model / GUI output is not a clinical diagnosis** and must not be used for medical decisions.

8. Common beginner mistakes

Mistake	Better habit
Heavy filtering on BMID _adi	Use BMID pass-through
Judging only by bright colors	Also check GT distance, confidence notes, Compare scores
Changing wave speed first	Fix angle/flip/arc before lowering c
Ignoring Log messages	Log explains skips, failures, and candidate notes
Thinking Yes candidate = proven tumor	It is a heuristic label for demos/learning

9. Mini glossary

- **S21** — transmission measurement between antennas.
- **Beamformer (DAS/DMAS/DMAS-D4)** — algorithm that turns many signals into one image.
- **ROI** — region of interest (the box around the hot spot).
- **GT** — ground truth tumor location from dataset metadata.
- **SCR / SNR** — how strong the peak is versus clutter/noise.
- **Pass-through** — skip cleaning because data is already cleaned.

10. Where files are saved

File	Meaning
results/latest_session_report.md	Full text report of last preprocess+reconstruct
results/latest_session_report_*.png	Selected / compare / ROI refine pictures
results/latest_guide_progress_report.md	Auto Validation guide report
results/latest_beginner_explainer.md	Auto Validation beginner report
This PDF	results/Beginner_Friendly_GUI_Guide.pdf